CAPSTONE PROJECT REPORT

The goal of this project was to build a model that predicts the musical genre of a song using its acoustic features. The instructions required a dimensionality reduction step, a clustering step, and a classifier. I also had to split the data by taking exactly 500 songs from each genre for the test set, and I set the seed using my NYU N-number (14418269) so that the results are reproducible.

Before training the model, I cleaned and prepared the dataset. Some features, such as duration or acousticness, were missing for a few songs, so I used median imputation to fill these gaps because the features are not normally distributed. The musical key was originally stored as text such as “C,” “F#,” or “A#,” so I mapped each key to a number from 0 to 11. The mode variable (“Major” or “Minor”) was converted to 1 and 0. Genre labels were turned into numbers using LabelEncoder. One genre appeared only 5 times in the dataset, which made the required train/test split impossible. I removed this genre to keep everything consistent. After this step, each of the 10 remaining genres contained 5000 songs. I randomly selected 500 songs per genre for the test set and kept the remaining 4500 for training, as required.

To reduce the dimensionality of the data, I used Linear Discriminant Analysis (LDA). LDA is a supervised method, which means it uses the genre labels to find directions that best separate the classes. This makes it more suitable for classification than PCA, which does not use label information. LDA reduced the original 14 features down to 9 new component. I fit LDA only on the training data and then applied the same transformation to the test data, which prevents leakage.

After LDA, I applied K-Means clustering with 10 clusters to the reduced training data. This clustering acted as the “unsupervised step” required by the assignment. However, when I plotted the K-Means results in the LDA space, the clusters were not very clear. The plot showed a lot of overlap between genres, which suggests that even though LDA helps with classification, it does not always create clean visual cluster boundaries.

When I visualized the data using Linear Discriminant Analysis (LDA), I expected the classes to appear more separated because LDA is a supervised dimensionality reduction method that uses the genre labels directly. The plot did show some structure—for example, a few genres formed tighter color regions—but overall there was still heavy overlap between most genres. This suggests that the acoustic features in the dataset do not naturally split into clean linear boundaries, even when the method is optimized to separate classes. I also tried applying K-Means clustering separately on the scaled features to see whether an unsupervised method could uncover clearer groups. However, the resulting visualization showed that the clusters were still broad and blended into each other, and the clusters did not match the true genres very well. This reinforces the idea that genre is not cleanly defined by a single set of acoustic properties and that simple clustering on linear feature spaces cannot capture the real underlying structure. These

observations also helped justify using t-SNE later in the project for visualization, since t-SNE is better at revealing nonlinear relationships that LDA and K-Means alone could not show.

To better understand the structure of the data, I tried t-SNE as a separate visualization tool. t-SNE is a nonlinear dimensionality reduction method that is very good at revealing patterns for plotting, even though it is not used for modeling. When I ran t-SNE on the scaled training data and then applied K-Means to the t-SNE output, the clusters looked much more distinct. The visualization showed curved groups and pockets of points that made more sense visually. This confirmed that even though LDA is better for building the classification model, t-SNE does a better job at showing how songs are naturally grouped in two-dimensional space. I did not use t-SNE or the t-SNE clusters in the classifier; I only used them for visualization and for understanding the dataset. The classifier was trained only on the LDA components to avoid leakage and to stay within the assignment rules.

Finally, I trained a random forest using the LDA features. I evaluated the model using the required test set of 500 songs per genre. I computed the ROC curve using micro-averaging and calculated the AUC. My final AUC score was strong , which means that the model performs well at predicting the correct genre. The most important factor behind this success was that LDA used the class labels to build directions that made it easier to separate genres. This gave the classifier clean, meaningful features to learn from.

Even though K-Means was not very clear in LDA space, the additional t-SNE visualization helped me understand the true shape of the data. The extra plot showed that musical genres do not form perfect, isolated groups. Instead, they sit on smooth, blended structures where some genres transition into others. This explains why classification is difficult and why nonlinear visualizations reveal better patterns. Overall, this project shows how dimensionality reduction, clustering, and classification can be combined to understand and predict musical genres.

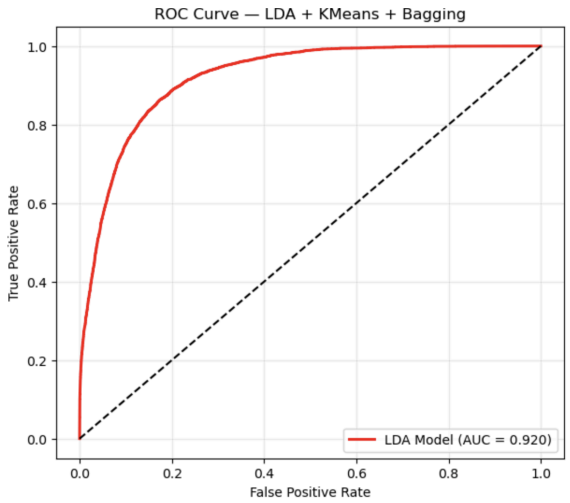
AUC - 0.9204308244444444

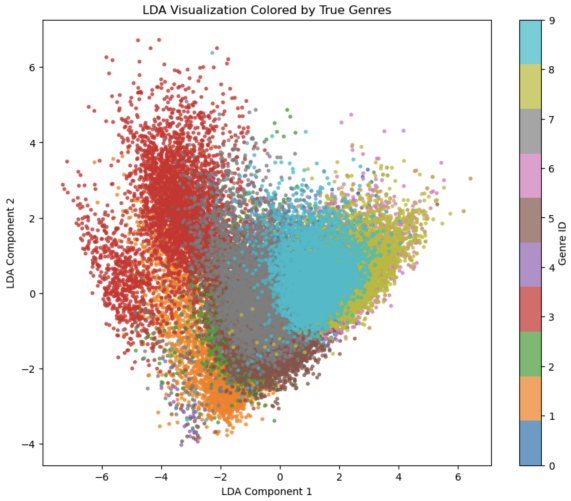
EXTRA CREDIT

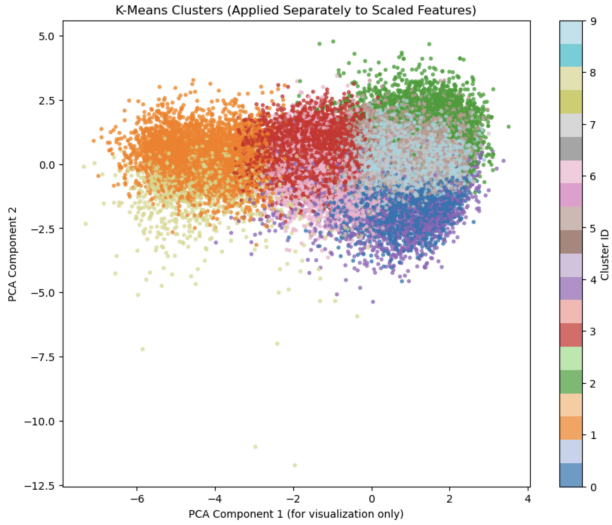
To better understand how similar the genres are in acoustic terms, I also computed the pairwise distances between their centroids in the LDA space. The results showed that some genres are extremely close together—for example, Genre 0 and Genre 4 had one of the smallest distances, indicating their acoustic profiles overlap heavily. In contrast, Genre 3 was consistently far from most other genres, suggesting it has a more unique sound signature. I also found that Genre 6 and Genre 8 were very close, which matched the visual overlap seen in the LDA plot. These distance patterns supported the idea that genre boundaries are gradual rather than sharply separated in this dataset.

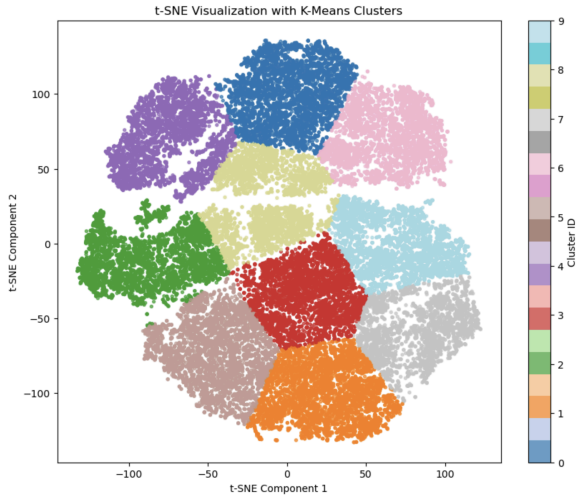
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Pairwise Distances Between Genre Centroids (LDA Space):
          Genre 0  Genre 1  Genre 2  Genre 3  Genre 4  Genre 5  Genre 6  \
Genre 0     0.000    2.888    1.970    4.440    1.101    1.938    2.127
Genre 1     2.888    0.000    1.638    3.488    2.644    2.160    4.459
Genre 2     1.970    1.638    0.000    3.520    1.433    2.189    3.555
Genre 3     4.440    3.488    3.520    0.000    4.288    4.208    5.639
Genre 4     1.101    2.644    1.433    4.288    0.000    2.246    2.721
Genre 5     1.938    2.160    2.189    4.208    2.246    0.000    2.940
Genre 6     2.127    4.459    3.555    5.639    2.721    2.940    0.000
Genre 7     1.974    2.267    1.549    3.255    1.822    1.900    3.068
Genre 8     1.996    4.501    3.569    5.653    2.638    2.975    0.365
Genre 9     1.179    3.824    2.677    4.846    1.563    2.883    2.369

          Genre 7  Genre 8  Genre 9
Genre 0     1.974    1.996    1.179
Genre 1     2.267    4.501    3.824
Genre 2     1.549    3.569    2.677
Genre 3     3.255    5.653    4.846
Genre 4     1.822    2.638    1.563
Genre 5     1.900    2.975    2.883
Genre 6     3.068    0.365    2.369
Genre 7     0.000    3.087    2.541
Genre 8     3.087    0.000    2.124
Genre 9     2.541    2.124    0.000
```






K-Means Clusters (Applied Separately to Scaled Features)
PCA Component 2
PCA Component 1 (for visualization only)
Cluster ID
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−6
−4
−2
0
2
4
9
8
7
6
5
4
3
2
1
0


t-SNE Visualization with K-Means Clusters
t-SNE Component 2
t-SNE Component 1
Cluster ID
100
50
0
−50
−100
−100
−50
0
50
100
9
8
7
6
5
4
3
2
1
0